Cancelling Translated Copies of Points in Convex Position

Progress Report

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Contents

1 The Problem

Definition 1. *A finite set S of points in $\mathbb{R}^2$ is in* convex position *if every point of S is a vertex of its convex hull* $\operatorname{conv}(S)$*.*

Definition 2 (Setup)**.** *Let S be a finite set of $n \geq 1$ points in $\mathbb{R}^2$ in convex position. Let T be a finite set of vectors in $\mathbb{R}^2$, each assigned a sign $\sigma(t) \in \{+1, -1\}$. For every $p \in \mathbb{R}^2$ define*

$$\sigma(p) \;=\; \sum_{\substack{t \in T \\ p-t \in S}} \sigma(t).$$

Write $\operatorname{supp}(\sigma) := \{p \in \mathbb{R}^2 : \sigma(p) \neq 0\}$*, and set* $T_\pm := \{t \in T : \sigma(t) = \pm 1\}$*.*

Theorem 1 (Lower Bound — proved)**.** $|\operatorname{supp}(\sigma)| \geq n$*.*

Theorem 2 (Main Theorem — partially proved)**.** *Assume $n \geq 3$ and T contains at least one positive and one negative vector. If $|\operatorname{supp}(\sigma)| = n$ then $n = 4$ and S is the vertex set of a parallelogram.*

Remark 1. *The condition "both signs present" is necessary: a single translate gives $|\operatorname{supp}(\sigma)| = n$ for any S. The theorem says: once genuine cancellation occurs but the support is still minimal, S is forced to be a parallelogram (and in particular $n = 4$).*

2 Proof Strategy

The proof is decomposed into six subproblems. Write $P := \operatorname{conv}(S)$, $Q := \operatorname{conv}(T)$. Label the vertices of P as $s_1, \ldots, s_n$ counterclockwise (indices mod n), with edge vectors $a_i := s_{i+1} - s_i$.

SP	Content	Status
1	Lonely Points Lemma + equality rigidity	**clean**
2	Exposed-translate theorem	**clean**
3	Boundary Minkowski structure	**clean**
4	Alternating-chain structure on each face	**clean**
5	No three independent edge directions	**open**
6	Two-direction case forces parallelogram	**clean**

How they combine. Subproblems 1–4 establish the rigid algebraic–geometric structure of any equality configuration. Subproblem 5 rules out three or more independent edge directions (the hard case). Subproblem 6 shows the remaining two-direction case forces $n = 4$ and a parallelogram. Everything is proved *except* Subproblem 5.

3 Subproblem 1: Lonely Points Lemma

Status: VERIFIED CLEAN — adversarial checker found no errors (3/3 independent runs).

Subproblem 1 (Lonely Points + Equality Rigidity)**.** Under the general setup (no equality hypothesis yet):

(a) **Lower bound:** $|\operatorname{supp}(\sigma)| \geq n$.

(b) **Equality rigidity:** If $|\operatorname{supp}(\sigma)| = n$, then:

(i) For each $s \in S$ there is a *unique* $t_s \in T$ such that $\ell_s := s + t_s$ is lonely (has a unique representation in $S + T$).

(ii) The lonely points ℓ_s are pairwise distinct and $\operatorname{supp}(\sigma) = \{\ell_s : s \in S\}$.

(iii) $\sigma(\ell_s) = \sigma(t_s) \in \{\pm 1\}$.

(iv) $\big||T_+| - |T_-|\big| \leq 1$.

Proof sketch

A point $p \in S + T$ is *lonely* if it has a unique representation $p = s + t$. For each $s \in S$, find a vector u_s that strictly exposes s in S (i.e. $\langle u_s, s\rangle > \langle u_s, s'\rangle$ for all $s' \neq s$; constructed by taking the nearest point in $\operatorname{conv}(S \setminus \{s\})$). Let t_s maximise $\langle u_s, \cdot\rangle$ on T. Then $\ell_s := s + t_s$ is lonely: any other representation $s' + t' = \ell_s$ would require $\langle u_s, s'\rangle + \langle u_s, t'\rangle = \langle u_s, s\rangle + \langle u_s, t_s\rangle$, forcing $s' = s$ and $t' = t_s$ by strict maximality. Since $\sigma(\ell_s) = \sigma(t_s) \neq 0$, we get n distinct support points.

For equality rigidity: if s had two lonely points $s + t$ and $s + t'$ with $t \neq t'$, we get $(n - 1) + 2 = n + 1$ distinct support points, a contradiction. The balance estimate follows from $\sum_p \sigma(p) = n(|T_+| - |T_-|)$ combined with $|\sum_p \sigma(p)| \leq n$. □

4 Subproblem 2: Exposed-Translate Theorem

Status: VERIFIED CLEAN — adversarial checker found no errors.

Under the equality hypothesis, write $t_i := t_{s_i}$ for the unique translate from SP 1 attached to vertex s_i. Define the normal cone

$$C_i := \{u \in \mathbb{R}^2 : s_i \text{ is the unique maximiser of } \langle u, \cdot\rangle \text{ on } S\}.$$

Subproblem 2 (Exposed-Translate Theorem)**.** For every $u \in C_i$, t_i is the unique maximiser of $\langle u, \cdot\rangle$ on T.

Proof sketch

If $u \in C_i$ and $t \in T$ maximises $\langle u, \cdot\rangle$ on T, the same exposed-maximum argument as in SP 1 shows $s_i + t$ is lonely. By SP 1 uniqueness, $t = t_i$. So $\operatorname{argmax}_T\langle u, \cdot\rangle = \{t_i\}$ for every $u \in C_i$. □

5 Subproblem 3: Boundary Minkowski Structure

Status: VERIFIED CLEAN — adversarial checker found no errors (3 runs).

Subproblem 3 (Boundary Minkowski Structure)**.** Under the equality hypothesis:

(1) The points $x_i := s_i + t_i$ are exactly the vertices of $R := P + Q$, in cyclic order.

(2) For each i, the exposed face of Q in the direction normal to the edge $[s_i, s_{i+1}]$ of P is the segment $[t_i, t_{i+1}]$, and

$$t_{i+1} - t_i = \lambda_i \, a_i \quad \text{for some } \lambda_i \geq 0.$$

Proof sketch

Let u_i be an outer normal to $[s_i, s_{i+1}]$. Small perturbations $u_i \pm \varepsilon a_i$ (for small $\varepsilon > 0$) enter cones C_i and C_{i+1} respectively; by SP 2, t_i and t_{i+1} are the unique maximisers of those perturbed directions on T. Letting $\varepsilon \to 0$, both t_i and t_{i+1} maximise $\langle u_i, \cdot \rangle$ on T, so both lie in the exposed face $G_i := F_Q(u_i)$. Since G_i is a compact convex segment parallel to a_i, and t_i, t_{i+1} are respectively the minimum and maximum of $\langle a_i, \cdot \rangle$ on G_i, we get $G_i = [t_i, t_{i+1}]$ by the interval structure of compact convex subsets of a line. That $\lambda_i \geq 0$ follows because x_i must be an endpoint (not interior) of the corresponding edge of R.

For part (1): the Minkowski-sum face formula $F_{P+Q}(u) = F_P(u) + F_Q(u)$ gives $F_R(u) = \{x_i\}$ for $u \in C_i$ (a vertex), and $F_R(u_i) = [x_i, x_{i+1}]$ (an edge). A cyclic-normal-order lemma (exterior angles sum to 2π) shows these are exactly all vertices and edges of R. $\square$

6 Subproblem 4: Alternating Chains on Faces

Status: VERIFIED CLEAN — adversarial checker found no errors.

For each i, let $F_i := [t_i, t_{i+1}] \cap T$ be the translates on the i-th face.

Subproblem 4 (Alternating-Chain Structure)**.** Under the equality hypothesis, for each i:

(1) There exists a non-negative integer m_i such that $t_{i+1} = t_i + m_i \, a_i$ and

$$F_i = \{t_i, \ t_i + a_i, \ t_i + 2a_i, \ \ldots, \ t_i + m_i \, a_i\}.$$

(2) The signs alternate along the chain: $\sigma(t_i + k \, a_i) = (-1)^k \, \sigma(t_i)$ for $0 \leq k \leq m_i$.

Proof sketch

Any interior point of edge $[x_i, x_{i+1}]$ of R has $\sigma = 0$ (by SP 1, the support equals $\{x_1, \ldots, x_n\}$, the vertex set of R). For $t = t_i + \lambda a_i \in F_i$ with $\lambda < m_i$, the point $p = s_{i+1} + t$ is interior to that edge and must cancel. The convex-position of S forces the only representations of p in $S + T$ to use translates t or $t + a_i$; since $\sigma(t) \neq 0$, the translate $t + a_i$ must also lie in T with opposite sign. This gives a successor rule $\lambda \in \Lambda \Rightarrow \lambda + 1 \in \Lambda$, and a predecessor rule $\lambda > 0 \Rightarrow \lambda - 1 \in \Lambda$. No fractional parameter can lie in Λ (it would give a lonely interior point with $\sigma \neq 0$), so $\Lambda \subset \mathbb{Z}$, and we conclude $\Lambda = \{0, 1, \ldots, m_i\}$ with alternating signs. □

7 Subproblem 5: No Three Independent Directions

Status: OPEN — This is the remaining hard case. Multiple attempts (versions v1–v19) have all contained critical errors. The problem is unsolved.

Subproblem 5 (No Three Independent Directions)**.** Under the equality hypothesis with the structure of SP 1–4, the edge vectors $a_1, \ldots, a_n$ of P span at most two unoriented directions (i.e. every a_i is a real multiple of one of two fixed vectors).

What is known

The following are proved and available.

1. **L^2-norm identity.** Since $\sigma(\ell_s)^2 = 1$:

$$n(|T|-1) \; = \underbrace{\sum_{\substack{t_1 \neq t_2 \\ \sigma(t_1)\sigma(t_2)=-1}} |S \cap (S + (t_1 - t_2))|}_{\text{opp-sign sum}} - \underbrace{\sum_{\substack{t_1 \neq t_2 \\ \sigma(t_1)\sigma(t_2)=+1}} |S \cap (S + (t_1 - t_2))|}_{\text{same-sign sum}} .$$

2. **Overlap bound.** For any $v \neq 0$, $|S \cap (S + v)| \leq 2$ (since S is in convex position).

3. **Coarse L^2 bound.** From (1) and (2):

$$n(|T| - 1) \; \leq \; 4\,|T_+|\,|T_-|.$$

   This gives $n \leq 4$ when $|T_+| = 1$ or $|T_-| = 1$, but *fails* for $|T_+| = |T_-| = k \geq 2$.

4. **Alternating chains** (SP 4): each face of P carries an arithmetic progression of translates with alternating signs.

Known dead-ends

- Assuming $|T| = 2$ is unjustified; explicit equality configurations with $|T_+| = |T_-| = 2$ exist (e.g. S a parallelogram, $T = \{0, a+b, -a, -b\}$).
- The support need not equal the vertex set of $\mathrm{conv}(S+T)$.
- Polynomial/lattice arguments do not apply in $\mathbb{R}^2$.
- No "opposite-sign lonely point" is guaranteed when all lonely signs are equal.

Most promising open approach

Lower-bound the same-sign sum to tighten the L^2 estimate. For $|T_+| = |T_-| = k \geq 2$ there are $2k(k-1)$ same-sign ordered pairs. If each contributes ≥ 1 to the same-sign sum, then

$$n(2k-1) \;\leq\; 4k^2 - 2k(k-1) \;=\; 2k^2 + 2k,$$

giving $n \leq 4$ for all $k \geq 2$. The needed ingredient is a proof that $|S \cap (S + (t_1 - t_2))| \geq 1$ for same-sign pairs (t_1, t_2), or an alternative argument using the alternating-chain structure from SP 4.

8 Subproblem 6: Two Directions Forces a Parallelogram

Status: VERIFIED CLEAN — adversarial checker found no errors (3/3 independent runs).

Subproblem 6 (Two Directions Forces a Parallelogram)**.** If S is in convex position with $n \geq 3$ and the edge vectors of $\mathrm{conv}(S)$ span at most two unoriented directions, then $n = 4$ and S is the vertex set of a parallelogram.

Proof

Step 1: Adjacent edges are non-parallel. If $e_i \parallel e_{i+1}$, then v_i, v_{i+1}, v_{i+2} are collinear, one being a strict convex combination of the other two. This contradicts convex position. So the two direction classes alternate around the polygon: $[u], [w], [u], [w], \ldots$, forcing n even.

Step 2: $n \leq 4$**.** The exterior angles $\varepsilon_i > 0$ sum to 2π, so the n directed edge directions $e_1, \ldots, e_n$ are pairwise distinct (their lifted arguments $\theta_1 < \cdots < \theta_n < \theta_1 + 2\pi$ are all in the same period). Since there are only two unoriented directions, there are at most four possible oriented directions $u, w, -u, -w$. Hence $n \leq 4$.

Step 3: $n = 4$**, parallelogram.** With n even, $n \geq 3$, $n \leq 4$, we get $n = 4$. Write the four edges as $B - A = au$, $C - B = bw$, $D - C = \sigma cu$, $A - D = \tau dw$ (with $a, b, c, d > 0$, $\sigma, \tau \in \{\pm 1\}$). The closure condition $(a + \sigma c)u + (b + \tau d)w = 0$ and linear independence of u, w force $\sigma = \tau = -1$, $a = c$, $b = d$. So opposite sides are parallel and equal: $ABCD$ is a parallelogram. □

9 Summary

The main theorem reduces entirely to Subproblem 5. All other ingredients are proved and verified.

Theorem 3 (Conditional)**.** *If Subproblem 5 holds, then Theorem 2 holds in full: every equality configuration with $n \geq 3$ and both signs in T forces $n = 4$ and S a parallelogram.*

Proof. SP 5 provides at most two edge directions; SP 6 then gives $n = 4$ and a parallelogram. □

What remains.

- Prove Subproblem 5.
- The key gap is the case $|T_+| = |T_-| = k \geq 2$; the L^2 bound alone is insufficient here.
- A lower bound $|S \cap (S + (t_1 - t_2))| \geq 1$ for same-sign pairs, combined with the L^2 identity, would complete the proof.
- See `proof/feedback_round5.txt` for a full account of failed approaches and a precise statement of the key missing lemma.

File guide.

`proof/section_1_lonely_rigidity_sol_v3.tex`	SP 1 (clean)
`proof/section_2_exposed_translate_sol_v1.tex`	SP 2 (clean)
`proof/section_3_boundary_minkowski_sol_v5.tex`	SP 3 (clean)
`proof/section_4_alternating_chains_sol_v2.tex`	SP 4 (clean)
`proof/section_5_three_directions.tex`	SP 5 problem statement
`proof/feedback_round5.txt`	SP 5 failure diagnosis
`proof/section_6_two_directions_sol_v3.tex`	SP 6 (clean)